Flower Type Classification - Level 1 Summary

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This project focuses on the classification of five flower species using deep learning techniques, with the goal of understanding how modern computer vision models extract meaningful patterns from complex visual data. Instead of relying on traditional rule-based approaches, the project follows a structured progression through increasingly sophisticated models, beginning with simple baselines and moving toward more advanced architectures. The dataset presents a challenging problem due to high intra-class variation and strong visual similarity across classes, which makes it difficult for models to distinguish between categories. By building and evaluating different models, this project aims to analyze how well each approach handles these challenges. Thus, the work serves as both a practical implementation of image classification and a study of how representation learning evolves across model complexity.

The main goal of our Level 1 goal is to establish a baseline for the classification task by implementing simple neural network models. This involves constructing a complete pipeline, starting with data loading and preprocessing, and extending to model training and evaluation. The intention is not to achieve optimal performance, but to observe how basic architectures behave when applied to high-dimensional image data. These initial results provide a reference point for future improvements, particularly when transitioning to more advanced techniques such as convolutional networks and transfer learning.

To briefly outline the methodology used in Level 1, the goal was structured as a series of three experiments, each meant to increase model complexity while maintaining a baseline. All experiments operated on the same dataset of five flower classes, with image resolution of 224x224 pixels with three color channels. The data pipeline included batching (batch size of 32), normalization through rescaling, and a standard train-validation split. The first experiment implemented a simple linear model by flattening the image into a one-dimensional vector and applying a fully connected layer with a softmax output. The second experiment extended this architecture by introducing a hidden layer with ReLU activation, allowing the model to learn non-linear relationships. The third experiment focused on hyperparameter tuning, incorporating techniques such as dropout, regularization, and variations in learning rate, batch size, and hidden layer size. Across all experiments, models were trained using the Adam optimizer and evaluated using cross-entropy loss and classification accuracy to ensure consistency in comparison. The sample of the dataset and results can be seen in Figure 1.

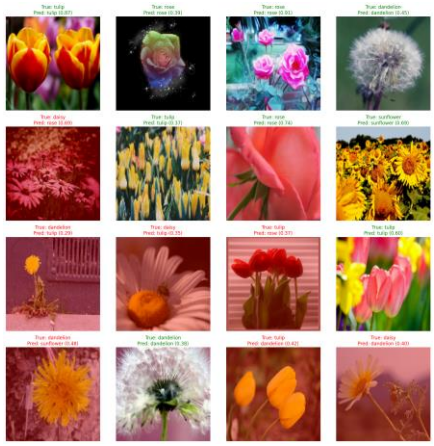

Figure 1. Flower Classification; incorrect predictions are shaded red.

In the first experiment, the linear model achieved a validation accuracy of approximately 45%, which is above the 20% baseline expected from random guessing in a five-class problem. However, this performance came with clear signs of overfitting, as training accuracy exceeded validation accuracy and the loss remained relatively high, seen in Figure 2. This suggests that while the model was able to capture some global patterns in the data, it lacked the ability to generalize effectively to unseen examples.

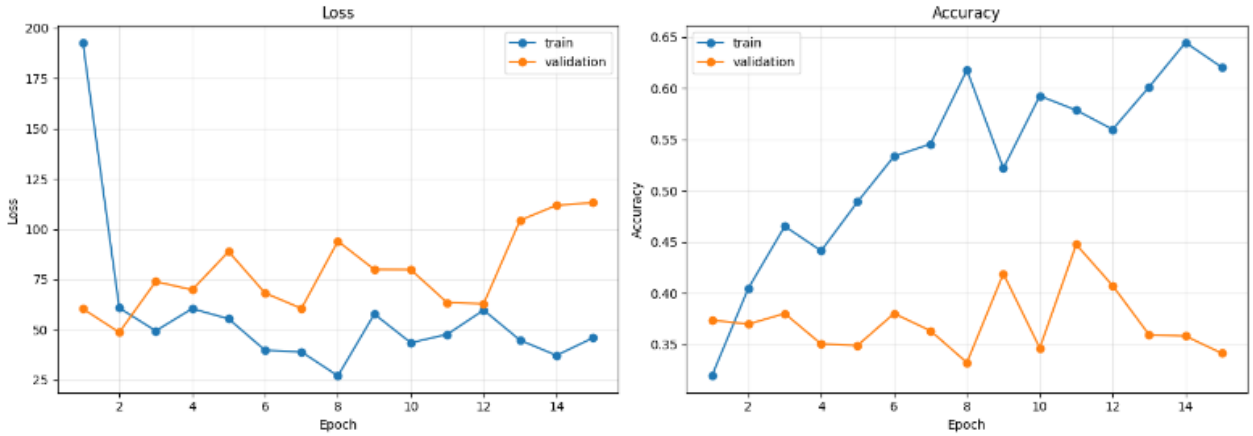

Figure 2. Experiment 1 – single flattened layer, fully connected neural network.

Introducing non-linearity in the second experiment led to improvements in how the model learned, especially in the reduction of loss values from extremely large magnitudes to a more stable range. This indicates that the model was better able to approximate underlying patterns in the data. However, this improvement did not translate into higher validation accuracy, which remained around 42%. As seen in Figure 3, the persistence of a gap between training and validation performance shows that expressiveness alone is not enough without an appropriate architectural design that accounts for spatial structure.

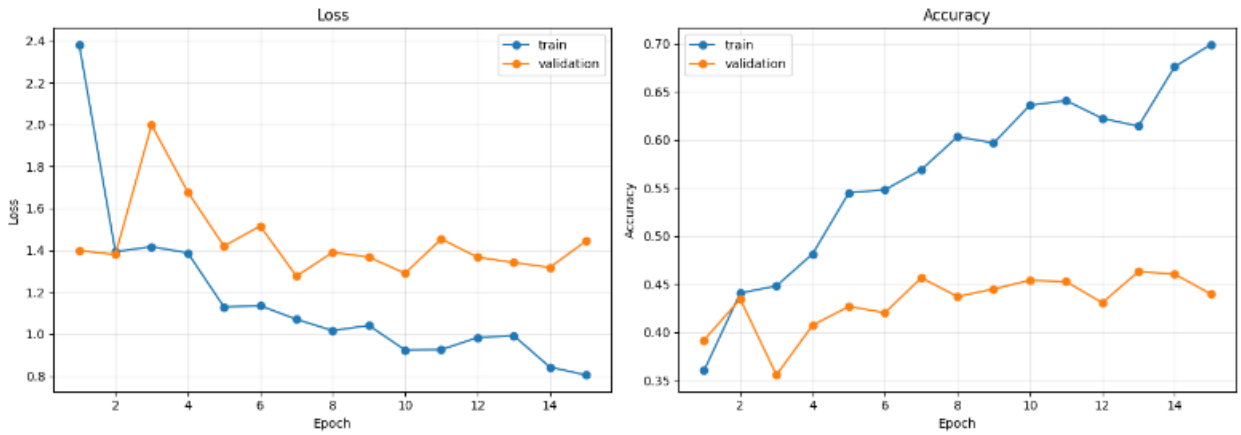

Figure 3. Experiment 2 – shallow fully connected network with ReLU activation.

The third experiment, which involved extensive hyperparameter tuning, produced the best results among the three, with validation accuracy reaching approximately 47% and peaking near 49.5% under optimal configurations. While this demonstrates that careful tuning can yield good improvements, the gains were relatively modest and appeared to plateau. This saturation indicates that the limitation is not simply the choice of hyperparameters, but rather the inherent constraints of fully connected architectures when applied to image data.

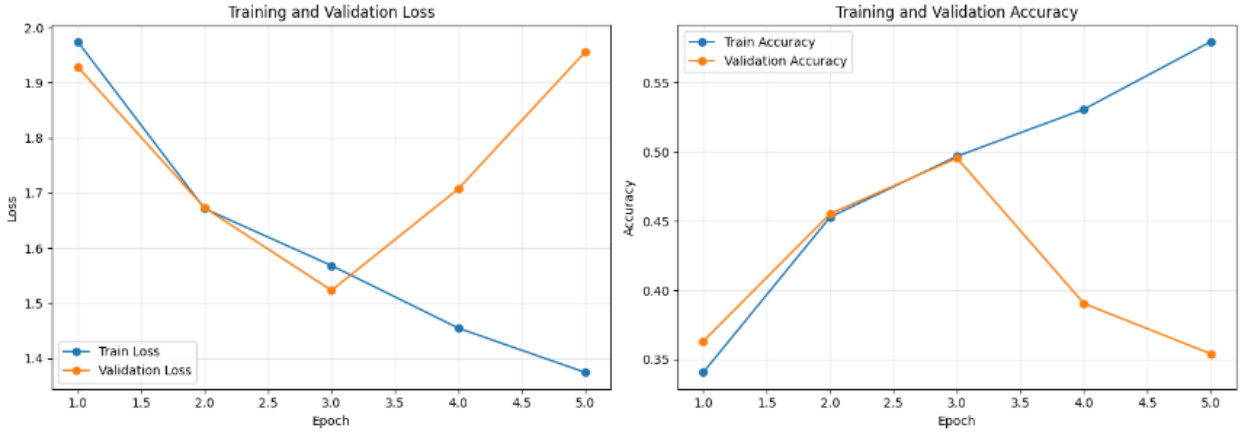

Figure 4. Experiment 3 – hyperparameter tuning.

In general, Level 1 results show a consistent limitation: while simple neural networks can outperform random guessing and learn basic patterns, they are fundamentally constrained. Even with added non-linearity and hyperparameter tuning, the models begin to reach a saturation point where further improvements are minimal. This suggests that learning from scratch with fully connected architecture is not sufficient for this task. As a result, the next stage of the project will shift toward transfer learning, specifically using pretrained models to utilize learned feature embeddings. By fine-tuning these models on the flower dataset and applying differential learning strategies, we hope to overcome the limitations observed in Level 1 and achieve better overall performance.